Politécnico do Porto
Instituto Superior de Engenharia do Porto

Development of an Integrated System for Acoustic Calibration of Stethoscopes

Rodrigo Pires

LEEC
Degree in Electrical and Computer Engineering

ISEP INSTITUTO SUPERIOR DE ENGENHARIA DO PORTO

Departamento de Engenharia Eletrotécnica
Instituto Superior de Engenharia do Porto

April, 2026

This report partially satisfies the requirements defined for the Project/Internship course, in the 3rd year, of the Degree in Electrical and Computer Engineering.

Candidate: Rodrigo Pires, No. 1210697, 1210697@isep.ipp.pt

Scientific Guidance: Luis Pinto Coelho, lfc@isep.ipp.pt

Departamento de Engenharia Eletrotécnica
Instituto Superior de Engenharia do Porto
Rua Dr. António Bernardino de Almeida, 431, 4200–072 Porto

April, 2026

Abstract

Aqui deverá ser apresentado o resumo de todo o trabalho efetuado. Esta secção não deverá exceder uma página.

Deve contextualizar o problema que pretende resolver ou a hipótese que irá formular, procure evidenciar as vantagens e desvantagens (se as houver) da solução encontrada, como também a forma através da qual a solução/hipótese foi validada. Neste último ponto, deverá referir-se aos desenvolvimentos efetuados, e à forma como validou (conformidade) e avaliou (desempenho) a solução encontrada.

O documento deve conter sempre duas versões do resumo: uma primeira no idioma do texto principal e a segunda num outro idioma. Este *template* assume que os dois idiomas em consideração são sempre Português e Inglês, assim, a classe irá colocar os cabeçalhos respetivos de acordo com o idioma selecionado nas opções da classe no ficheiro `main.tex`.

Keywords: Lista, separada por vírgulas, de palavras, frases, ou acrónimos chave no âmbito do trabalho descrito neste texto.

Resumo

The summary of all the developed work should be presented here. This section should not exceed one page.

Start the abstract with the contextualization of the problem you intend to solve or the hypothesis you will formulate. Try to highlight the advantages and disadvantages (if any) of the solution found, as well as the way in which the solution/hypothesis was validated. In this last point, you should refer to the developments made, and to the way you validated (compliance) and evaluated (performance) the solution found.

The document must always contain two versions of the abstract: a first in the language of the main text and the second one in another language. This template assumes that the two languages are always Portuguese and English, therefore, the class will place the correct section headers according to the language selected in the class options in the `main.tex` file.

Palavras-Chave: Comma separated list of words, phrases, or key acronyms within the scope of your developed work.

Contents

List of Figures **vii**

List of Tables **ix**

Listings **xi**

List of Algorithms **xiii**

List of Symbols **xv**

1 Introduction **1**

1.1 State of the Art/Contextualization 1

1.2 Project Description 2

1.2.1 Objectives 2

2 Methodology **3**

2.1 Literature Review 3

2.1.1 Sound Generation Technologies 3

References **5**

List of Figures

List of Tables

Listings

List of Algorithms

List of Symbols

Symbol	Description	Units
f	força	N
i	corrente	A
M	massa	kg
P	potência	W
x	deslocamento	m
θ	deslocamento angular	rad
ω	velocidade angular	$\mathrm{rad\,s^{-1}}$

Chapter 1

Introduction

1.1 State of the Art/Contextualization

Stethoscopes remain the most fundamental and prevasive diagnostic tools in clinical practice for the auscultation of cardiopulmonary and vascular sounds. Currently, traditional acoustic stethoscopes dominate over 90% of the global market due to their high durability, low cost, and independence from electrical power sources [1]. However, purely acoustic auscultation is inherently limited not only by the fallibility of human auditory perception but also by the unrecognized physical degradation of the instrument over time. Studies have demonstrated that general practitioners miss up to 57% of significant valvular heart diseases when relying on standard stethoscopes [2].

While frequently attributed to human auditory limitations, this high error rate may be significantly composed by unnoticed material failures; such as compromised acoustic seals in the eartips, hardened tubing, or worn diaphragms. These physical degradations silently alter the stethoscope's frequency response, attenuating vital low-frequency biological sounds before they ever reach the clinician's ear. To mitigate these human diagnostic errors, the medical field is experiencing the gradual emergence of electronic stethoscopes designed to amplify, filter, and digitize biological sounds.

This technological shift, while clinically beneficial, introduces a critical new engineering problem: these electronic devices actively alter and process the raw audio

signal. Because both the acoustic integrity of traditional tubing and the digital processing of modern devices directly impact diagnostic accuracy, the development of a precise calibration system to verify signal fidelity has never been more urgent.

1.2 Project Description

This project proposes the development of an acoustic calibration system designed to characterize the frequency response, and signal fidelity of stethoscopes. The proposed architecture consists of two primary domains: a physical hardware module for sound generation and detection, and a software platform for control, data acquisition, and spectral analysis.

1.2.1 Objectives

By the end of the project, the developed system should be able to:

- **Simulate Cardiopulmonary Acoustics:** Generate controlled acoustic signals that mimic the frequency spectrum of biological sounds, particularly in the frequency range for cardiopulmonary auscultation.
- **Capture High-Fidelity Audio Data:** Accurately measure the stethoscope's output signal using a high-fidelity microphone and data acquisition system, ensuring that the measurement setup itself does not introduce significant distortions.
- **Analyze Frequency Response:** Analyze the acquired audio data to compute the stethoscope's frequency response curve utilizing python-based spectral analysis techniques.
- **Establish a Repeatable Protocol:** Develop a standardized testing protocol that can be easily replicated for routine stethoscope calibration in clinical settings, ensuring consistent and reliable performance assessments.

Chapter 2

Methodology

2.1 Literature Review

First and foremost, for this acoustic calibration of stethoscopes it was needed to, understand in which frequencies the cardiopulmonary sounds operate in. This is important to know which frequencies we need to be able to reproduce in our calibration system, and also measure the stethoscope's response.

According to studies, the frequency range of heart sounds is typically between 5 Hz and 600 Hz, while lung sounds can range from 100 Hz to 2000 Hz [4]. Other studies have also shown that the most important frequencies for heart sounds and diseases are typically between 20 Hz and 150 Hz, while for lung sounds, the most important frequencies are between 100 Hz and 1000 Hz [5]. This means that our calibration system should prioritize the ability to reproduce sounds in these frequency ranges, as they are the most relevant for clinical auscultation.

2.1.1 Sound Generation Technologies

Traditional audio testing frequently relies on standard dynamic speakers. However, dynamic speakers operate by displacing free air.

Because a medical stethoscope is clinically designed to capture vibrations through high-impedance solid mediums (such as human tissue and fluid), utilizing an air-coupled speaker introduces significant energy loss, particularly in the low-frequency cardiovascular bands (20 Hz - 150 Hz).

To overcome this limitation, modern biomedical testing methodologies require the use of surface transducers or exciters. Instead of pushing air, surface transducers are designed to directly vibrate the solid object to which they are attached. When a surface transducer is mechanically coupled with a tissue-simulating medium (such as silicone or acoustic gelatine), it can accurately replicate the acoustic impedance of a human chest [6].

ollowing the justification for structure-borne sound converters, three commercially available surface transducers were selected for evaluation in this project:

- **Dayton Audio TT25-8 PUCK:** This is a compact, high-performance tactile transducer designed for low-frequency applications. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 19€.
- **Dayton Audio HDN-8 Weatherproof:** This is a high-performance, weatherproof tactile transducer designed for outdoor use. It can produce strong vibrations in the 20 Hz to 80 Hz range and comes at a price range of around 25€.
- **Dayton Audio DAEX25FHE-4:** This is a compact and affordable exciter with a frequency response of 70 Hz to 20 kHz and comes at a price range of around 15€.

References

[1] Transparency Market Research, "Stethoscope market - global industry analysis, size, share, growth, trends, and forecast, 2022-2031." Industry Report, 2022. [Online]. Available: https://www.transparencymarketresearch.com/. [Cited on page 1]

[2] S. K. M. Gardezi, S. G. Myerson, B. Prendergast, A. Prothero, A. Prendergast, M. Loudon, and J. D'Arcy, "Cardiac auscultation in diagnosing valvular heart disease: a comparison between general practitioners and cardiologists," *European Heart Journal*, vol. 38, no. suppl_1, 2017. [Cited on page 1]

[3] S. D. Poonguzhali, T. N. Prabakar, and V. Elamaran, "Fpga-based design and implementation of cascaded digital filters for enhanced auscultation in digital stethoscopes," *Circuits, Systems, and Signal Processing*, 2025.

[4] M. Tavakoli *et al.*, "Heart and lung sound measurement using an esophageal stethoscope with adaptive noise cancellation," *Sensors*, vol. 21, no. 20, p. 6757, 2021. [Cited on page 3]

[5] J. K. Roy, T. S. Roy, and S. C. Mukhopadhyay, "Heart sound: Detection and analytical approach towards diseases," in *Modern Sensing Technologies*, pp. 117–139, Springer, 2019. [Cited on page 3]

[6] M. Klum *et al.*, "Validation system for digital stethoscopes," *Current Directions in Biomedical Engineering*, vol. 5, no. 1, pp. 261–264, 2019. [Cited on page 4]